CSE221 Assignment 01 Spring 2026

A. Odd or Even?

1 second, 256 megabytes

Forbidden Keywords for the Quiz: sort, open, file, creat(, fstream, thread, process, system(, exec(

Do you know how to tell if a number is **Odd** or **Even**? You are given T numbers, and for each of those numbers, you have to tell whether the number is odd or even.

Input

The first line will contain a single integer T ($1 \leq T \leq 100$). Each of the next T lines will contain a number N ($-10^5 \leq N \leq 10^5$).

Output

For each N, you have to print whether the number is odd or even. See the sample input-output format to know what exactly you have to print.

input

```
5
10
19
7
3
100
```

output

```
10 is an Even number.
19 is an Odd number.
7 is an Odd number.
3 is an Odd number.
100 is an Even number.
```

B. Can you solve Arithmetic Expressions?

1 second, 256 megabytes

Forbidden Keywords for the Quiz: eval, sort, open, file, creat(, fstream, thread, process, system(, exec(

Can you solve arithmetic expressions with your programming knowledge? Let's find it out. You will be given some arithmetic expressions, and you have to solve them.

Input

The first line will contain a number $T(1 \leq T \leq 1000)$ representing the number of test cases. Then for each test case, you will be given an arithmetic expression. Please see the sample input below. It is guaranteed that the numbers inside the arithmetic expression will be between 1 and 1000.

Output

For each test case, you have to print the result. Look at the sample output for reference.

Important Note: Your answer might contain floating point numbers, and in that case, your answer doesn't have to be exactly equal to the actual answer. For example, if your answer is 20.250000001 and the judge's solution is 20.25, your answer will still be considered correct. As long as it is really close to the correct solution, your solution will be considered correct. Formally speaking, if your solution is x, and the judge's solution is y, then as long as $|x - y| \leq 10^{-6}$, your solution will be correct. In the above example, your solution was 20.250000001 and the judge's solution was 20.25. If you take the difference of these two numbers, they are smaller than 10^{-6}. Similarly, if the judge's solution is 19.0000000000 and your solution is 19, it is still correct, as the difference is 0, which is less than 10^{-6}.

input

```
15
calculate 67 + 41
calculate 85 / 5
calculate 13 - 56
calculate 99 - 95
calculate 3 / 10
calculate 12 * 19
calculate 14 - 6
calculate 3 * 88
calculate 45 * 68
calculate 81 - 0
calculate 77 + 40
calculate 8 * 84
calculate 73 - 22
calculate 85 - 86
calculate 28 * 58
```

output

```
108.000000
17.000000
-43.000000
4.000000
0.300000
228.000000
8.000000
264.000000
3060.000000
81.000000
117.000000
672.000000
51.000000
-1.000000
1624.000000
```

C. Fast Sum

1 second, 256 megabytes

Forbidden Keywords for the Quiz: sort, open, file, creat(, fstream, thread, process, system(, exec(

Your friend is trying to solve the following problem. You are given $\mathbf{T}$ test cases. For each test case, you are given an integer $\mathbf{N}$. You have to find out the summation of 1 to N. More formally, your friend has to calcuate

$$\sum_{x=1}^{x=N} x$$

Your friend wrote the following code in Python to solve it:

```
T = int(input())
```

```python
for _ in range(T):
    N = int(input())
    sum = 0
    for i in range(1, N + 1):
        sum += i
    print(sum)
```

Same code in Java:

```java
import java.util.Scanner;

public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int T = sc.nextInt();
        for (int t = 0; t < T; t++) {
            int N = sc.nextInt();
            int sum = 0;
            for (int i = 1; i <= N; i++)
                sum += i;
            System.out.println(sum);
        }
    }
}
```

However, the code is not passing the online judge due to some unknown errors for large values of N.

Since you are currently studying CSE221 and have learned about time complexity, help your friend come up with a more efficient solution.

Input

The first line contains a single integer T $(1 \le T \le 10^4)$ — the number of test cases.

The next T lines each contain a single integer N $(1 \le N \le 10^6)$.

Output

For each test case, print a single integer — the summation from 1 to N.

input

```
5
2
5
10
12
100
```

output

```
3
15
55
78
5050
```

D. Is Sorted?

1 second, 256 megabytes

Forbidden Keywords for the Quiz: sort, open, file, creat(, fstream, thread, process, system(, exec(

You are given an array of N integers. Determine whether the given array is in non-decreasing order.

An array is said to be in non-decreasing order if, for every valid index i such that $1 \le i < N$ (1-based indexing), the condition $A[i] \le A[i+1]$ holds true.

Example: $[1, 2, 4, 5]$, $[1, 2, 2, 4, 4, 5]$ are in the non-decreasing order because every element is less than or equal to the one after it.

Input

The first line contains a single integer T $(1 \le T \le 100)$ — the number of test cases. Each testcase contains two lines.

In each test case, the first line contains a single integer N $(1 \le N \le 10^4)$ — the number of elements in the array. The second line contains N integers separated by spaces $a_1, a_2, a_3 \ldots a_n$ $(1 \le a_i \le 10^6)$ — the elements of the array.

Output

For each test case, print `YES` if the array is in non-decreasing order. Otherwise, print `NO`.

input

```
3
4
1 2 3 3
4
1 5 2 6
1
5
```

output

```
YES
NO
YES
```

E. Reverse Sorting

2 seconds, 256 megabytes

Forbidden Keywords for the Quiz: sort, open, file, creat(, fstream, thread, process, system(, exec(

You are given an array A of $\mathbf{N}$ integers. Your task is to sort the array in non-decreasing order using only a specific type of operation:

- In one operation, you may select any subarray of length exactly 3 and reverse it. In other words, you choose some $1 \le i, j \le N$ such that $j - i + 1 = 3$ and reverse `A[i...j]`.

You can apply this operation as many times as you like (or not at all). Your goal is to determine whether it is possible to sort the array using only this operation. If it is possible, you have to print the operations too. Look at the output format for a better understanding.

Input

The first line contains an integer N $(1 \le N \le 1000)$. The second line contains N integers A_i $(1 \le S_i \le 10^5)$.

Output

If it is not possible to make the array non-decreasing, print **NO**. Otherwise print **YES**. If your answer is **YES**, you have to output the number of moves you needed, let it be M. Then the next M lines will contain pairs of (i, j) representing a valid operation. Make sure your moves are valid, and those moves make the array A non-decreasing.

NOTE: If you don't understand the output format properly, look at the sample input-output and explanation.

input

```
4
2 3 1 1
```

output

```
YES
2
1 3
2 4
```

input

```
6
2 5 5 1 5 5
```

output

```
NO
```

input

```
1
2
```

output

```
YES
0
```

input

```
2
6 6
```

output

```
YES
0
```

For the Sample Input, one way to sort the array $[2, 3, 1, 1]$ using only the allowed operation:

1. Choose indices $(1, 3)$ and reverse the subarray $[2, 3, 1]$ which becomes $[1, 3, 2]$. So the resulting array: $[1, 3, 2, 1]$
2. Choose indices $(2, 4)$ and reverse subarray $[3, 2, 1]$ which becomes $[1, 2, 3]$. So the resulting array: $[1, 1, 2, 3]$

F. An Ancient Sorting Algorithm

1 second, 256 megabytes

Forbidden Keywords for the Quiz: sort, open, file, creat(, fstream, thread, process, system(, exec(

You are given an array of N integers. You have to sort the array in non-decreasing order using a custom sorting algorithm with the following constraint:

- You may only swap adjacent elements with the same parity (i.e., both even or both odd)

Sort the array in non-decreasing order until no more such swaps are possible and print the final array.

Input

The first line contains a single integer N $(1 \leq N \leq 1000)$ — the number of elements in the array.

The second line contains N integers separated by spaces $a_1, a_2, a_3 \ldots a_n$ $(1 \leq a_i \leq 10^6)$ — the elements of the array.

Output

Print the final array after sorting the array in non-decreasing order until no more such swaps are possible.

input

```
7
4 2 4 7 1 6 1
```

output

```
2 4 4 1 7 6 1
```

input

```
5
3 5 9 7 1
```

output

```
1 3 5 7 9
```

input

```
14
4 8 2 9 1 5 4 6 8 1 7 13 11 8
```

output

```
2 4 8 1 5 9 4 6 8 1 7 11 13 8
```

input

```
1
221
```

output

```
221
```

G. Sorting Again??

1 second, 256 megabytes

Forbidden Keywords for the Quiz: sort, open, file, creat(, fstream, thread, process, system(, exec(

Suppose you are given a task to rank the students. You have gotten the marks and ID of the students. Now your task is to rank the students based on their marks using a sorting algorithm. If two or more students get the same mark, then students with the lower ID will get prioritized. See the input and output for a better understanding.

However, **you have to keep in mind that your sorting algorithms perform the minimum number of swapping operations.**

Input

The first line contains an integer T $(1 \leq T \leq 100)$ denoting the number of test cases.

For each test case:

The first line contains an integer N $(1 \leq N \leq 1000)$.

The second line contains N distinct integers S_i $(1 \leq S_i \leq 1000)$, where S_i denotes the student ID.

The third line contains N integers S_m $(1 \leq S_m \leq 1000)$, where S_m denotes the obtained mark of the corresponding student.

Note: It is guaranteed that all student IDs are unique.

Output

For each test case, the output must contain a number X which denotes the number of minimum swaps. The rest of the N lines will contain the Student ID and obtained marks sorted based on the instruction above. See the sample output for a better understanding.

Important Note: Since you are asked to minimize the number of swaps, if your number of swaps doesn't match with the judge's answer, your solution will be considered incorrect.

TO CHANGE: Look at the first testcase of the sample input. It can be shown that this can be sorted with only 4 swaps. It can also be shown that it is not possible to sort this in less than 4 swaps.

input

```
2
7
7 4 9 3 2 5 1
40 50 50 20 10 10 10
3
1 2 3
40 59 10
```

output

```
Minimum swaps: 4
ID: 4 Mark: 50
ID: 9 Mark: 50
ID: 7 Mark: 40
ID: 3 Mark: 20
ID: 1 Mark: 10
ID: 2 Mark: 10
ID: 5 Mark: 10
Minimum swaps: 1
ID: 2 Mark: 59
ID: 1 Mark: 40
ID: 3 Mark: 10
```

H. Trains 2

1 second, 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

Note: You can use the built-in sort function for this problem only for this problem in this lab.

You have recently been recruited as the Software Engineer at Jumanji Railway Software System. You have a big task at hand. You will be given the departure schedules for N trains. The next N lines will contain the name of the train, the destination, and the departure time.

Sort the given lines in the **special lexicographical order** based on the name of the trains. If two or more trains have the same name, then the train with the latest departure time will get prioritized. If there is still a tie, then the train which comes first in the input will come first.

Input

The first line will contain an integer N ($1 \le N \le 1000$). For the next N lines, the i-th line will that format the following format:

```
train_name will depart for destination at departure_time
```

The departure time will be given in 24-hour `hh:mm` format. It is guaranteed that `train_name` and `destination` will contain no whitespace characters, **only contain lowercase or uppercase Latin letters**, and never be longer than 50 characters.

Output

Print the N lines sorted according to the specified order.

input

```
6
HogwartsExpress will depart for Hogwarts at 11:15
AppleExpress will depart for AppleLand at 15:00
HogwartsExpress will depart for Ilvermony at 11:10
dhumketuexpress will depart for Rajshahi at 06:00
dhumketu will depart for Rajshahi at 06:00
AppleExpress will depart for OrangeLand at 15:00
```

output

```
dhumketu will depart for Rajshahi at 06:00
dhumketuexpress will depart for Rajshahi at 06:00
AppleExpress will depart for AppleLand at 15:00
AppleExpress will depart for OrangeLand at 15:00
HogwartsExpress will depart for Hogwarts at 11:15
HogwartsExpress will depart for Ilvermony at 11:10
```

The **special lexicographical order** is defined as using the following alphabetical ordering:

```
abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ
```

To put it simply:

- All lower case letters come first in normal alphabetical order.
- All the upper case letters come after that.

Let S and T be strings. We say that $S < T$ if one of the following is true:

- Let i be the first position where $S[i] \neq T[i]$. Then we say that $S < T$ if $S[i]$ comes before $T[i]$ in our alphabetical ordering.
- S is a proper prefix of T.

Examples:

- `"anaconda"` < `"apple"` because they are different at the second letter (`'n'` for `"anaconda"` and `'p'` for `"apple"`) and `'n'` comes before `'p'` in our alphabetical ordering.
- `"apple"` < `"Anaconda"` because they are different at the first letter, and `'a'` comes before `'A'` in the alphabetical ordering.
- `"apple"` < `"appleworld"` because `"appple"` is a proper prefix of `"appleworld"`. In other words, it is because `"appleworld"` starts with `"apple"`.

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